



BUSINESS BLUEPRINT

AOP AUTOMATION FOR DMD EOEG PLANT

Honeywell

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Business Blueprint

AOP Automation for DMD EOEG Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

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Revision History

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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for DMD EOEG plant for DMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various MEG plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for MEG technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
DMD EOEG DCT	-	Digital AOP DMD EOEG DCT	 Digital AOP DMD EOEG DCT.xlsx
UAT Observation list		Digital AOP MEG UAT Observation list	 Digital AOP MEG UAT Observations.xl
EOE effective Operating Hours calculation		RIL AOP - MEG Maintenance Screen - Slowdown Logic_R&T Validated	 RIL AOP - MEG Maintenance Screen

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

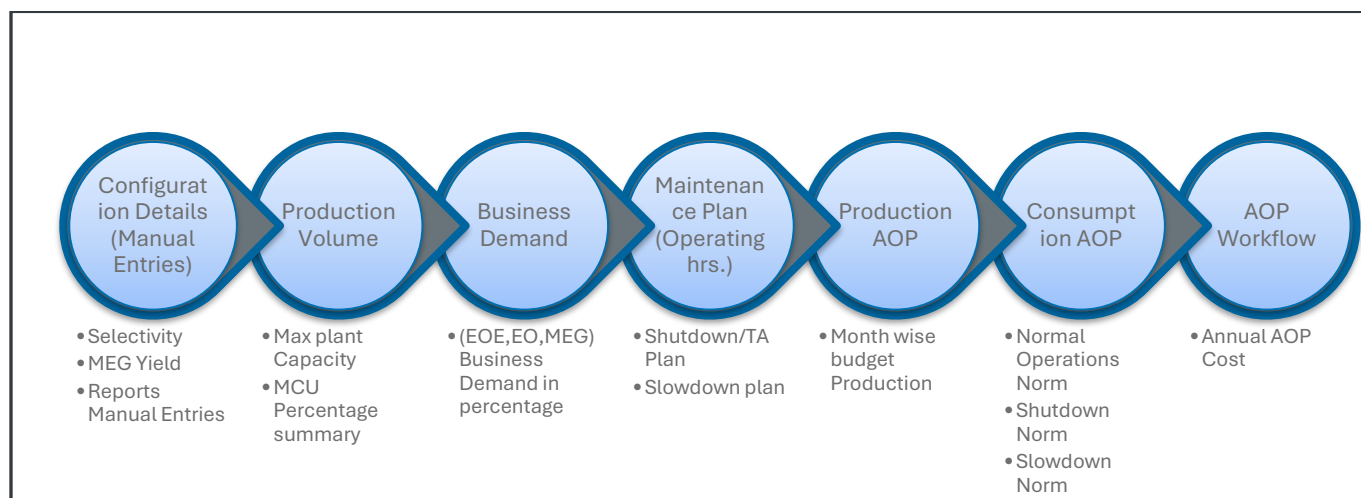


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Production AOP

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Configuration

The configuration screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data, including Yield, Selectivity, and Consumption, are configured under the Configuration screen to support Production and Consumption Norms calculations and reports generation.

By default, the values entered for the current year will be automatically carried forward to the next year.

The user must insert the TST value month wise and save it with a valid reason. This will provide the Latent heat against TST.

NOTE- Latent heat will be calculated only if the user provides the TST temperature; otherwise, the latent heat value will be set to 0.

In configuration screen, below additional features and tabs are provided.

- AOP historical period basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- Constant tab is provided to enter the constant value which will not change month wise as per below snapshot.
- Report manual entry tab is provided to enter report specific manual entries for last budgeted FY actual values (e.g. Budgeted selectivity, Actual Selectivity, guaranteed selectivity and actual operating Hours)
- Option to import and export manually entered from configuration screen is provided.

The screenshot below highlights the details captured in the digitized AOP platform.

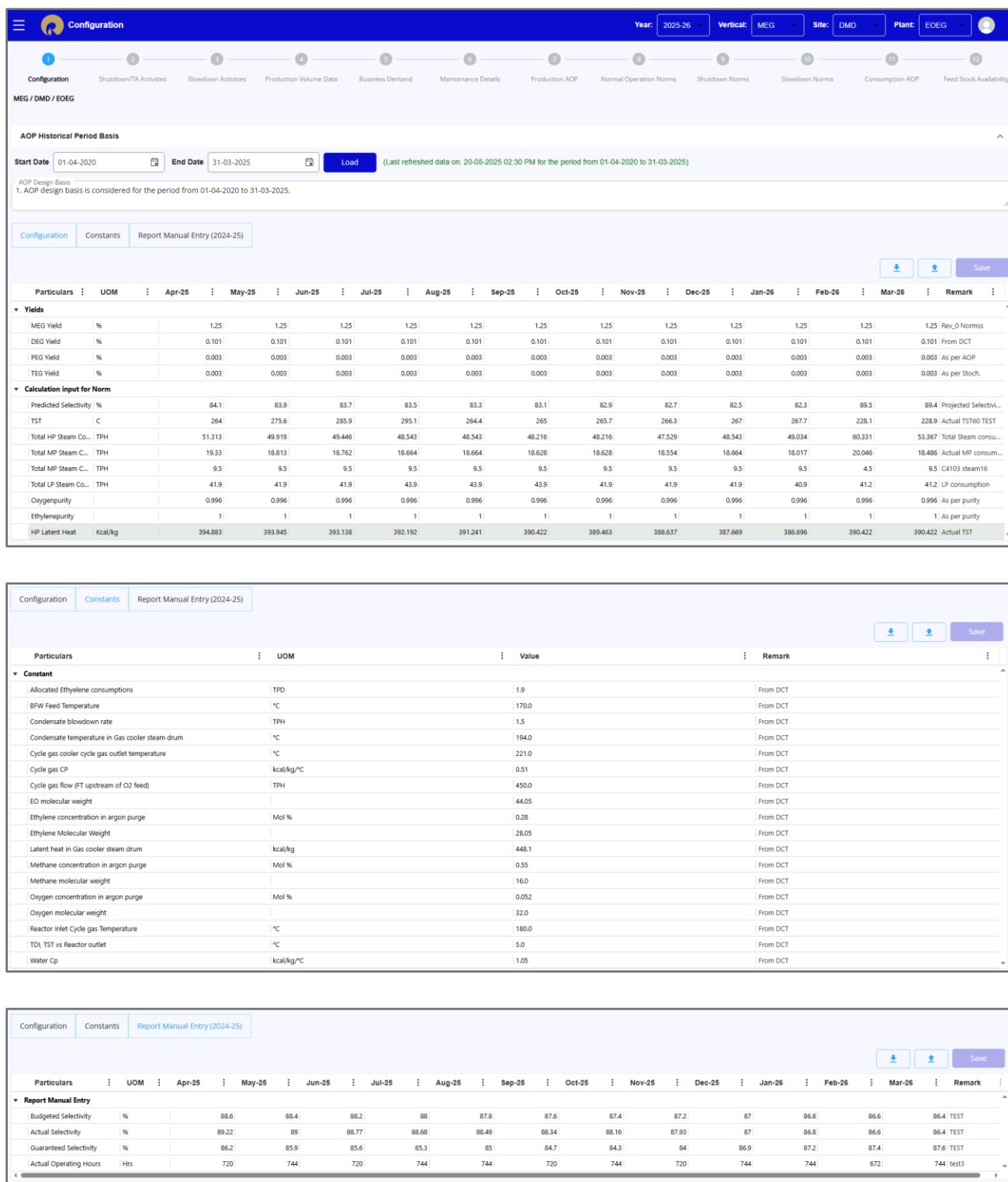


Figure 2 Digitized AOP screen for Configuration/manual Inputs

2.1.1.2 Production Volume Data

The Max Capacity (MC) of EOEG plants is defined on an EOE basis.

MC values for Ethylene Oxide (EO), Ethylene Oxide Equivalent (EOE) and Monoethylene Glycol (MEG) will be captured separately.

Month wise data is considered for deriving the best value. This will account for seasonal effects as well as the festive season.

If AOP basis is considered as 05 years, value estimation involves averaging production data over continuous 5-day periods in the last 5 years, after removing outliers and considering the normal operating mode without recycle for each month, 5 such averages are calculated and the maximum average value from the available 05 average set is designated as the MC value or production volume. Also, the edit option provided to change the derived MC value.

NOTE- AOP historical period basis added in configuration screen where user can change Start date and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

A similar approach is employed for calculating the EO MC value.

The following logic has been formulated to derive month-wise EOE and EO MC values.

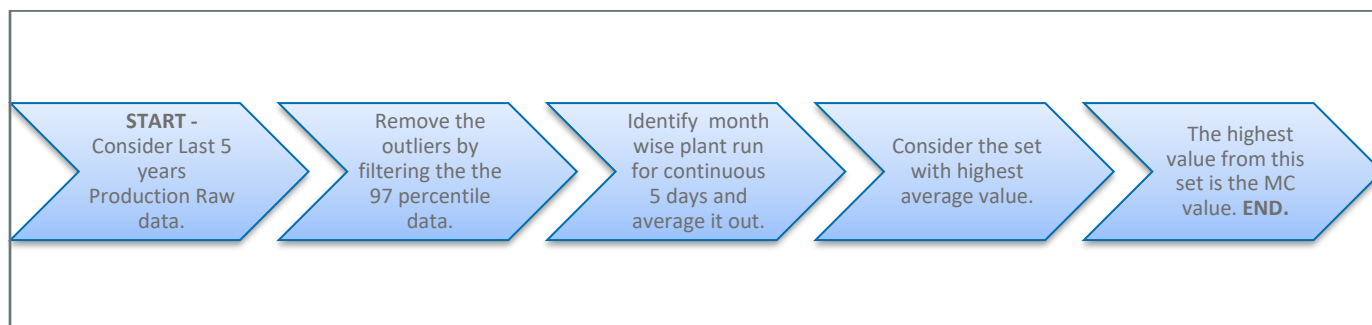


Figure 3 Digitized AOP Steps to derive EOE and EO month wise MC values

The MEG MC value is determined using the following formula.

$$MEG MC = MEG Yield * (EOE MC - EO MC)$$

Where MEG yield is a month-wise manual entry captured in the AOP digitization portal in the Configuration screen, as seen in the screenshot below.

Particu...	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	%	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287	1.287 / AOP FY25-26
PEG Yield	%	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24 / AOP FY25-26

Figure 4 Digitized AOP screen for MEG Yield

If the user needs to modify the production rate this can be done by providing a specific reason.

The AOP digitization portal captures a month-wise percentage summary, comparing each month's value against the highest MU value recorded over the past 12 months.

This information is captured in the AOP digitization portal as seen in the screenshot below.

Production Volume Data

Year:2025-26

Vertical:MEG

Site:OMD

Plant:EOEG

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10

11

12

Configuration

Shutdown/TA Activities

Slowdown Activities

Production Volume Data

Business Demand

Maintenance Details

Production AOP

Normal Operation Norms

Shutdown Norms

Slowdown Norms

Consumption AOP

Feed Stock Availability

MEG / DMD / EOEG

Download

Print

Save

Calculate

Select UOM
TPH

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
EOE	21.25	21.25	21.25	21.25	21.25	21.25	21.25	21.25	21.25	21.25	21.25	21.25	21.25 As per Extra O2 dive...
EO	9.236	9.259	9.531	9.577	9.43	9.53	9.554	9.059	9.028	9.584	9.552	9.457	Click to add remark
MEG	15.017	14.988	14.648	14.592	14.775	14.65	14.62	15.239	15.277	14.583	14.622	14.741	Click to add remark

Percentage Summary

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
EOE	100	100	100	100	100	100	100	100	100	100	100	100
EO	96.37	96.61	99.45	99.92	98.4	99.44	99.69	94.52	94.2	100	99.67	98.68
MEG	98.3	98.11	95.88	95.51	96.71	95.89	95.7	99.75	100	95.45	95.71	96.49

Figure 5 Digitized AOP screen for Production Volume data

The '**Calculate**' button on this screen allows the user to compute the MEG MC value if any changes have been made to the MEG Yield and Start and End date for AOP basis selection on the configuration screen.

2.1.1.3 Business Demand

The production demand is considered as the maximum value from last year's achieved production values. The default value for EOE, EO and MEG is 100%.

The business demand value can be modified by users upon providing a specific reason.

This information is captured in the AOP digitization portal as seen in the screenshot below.

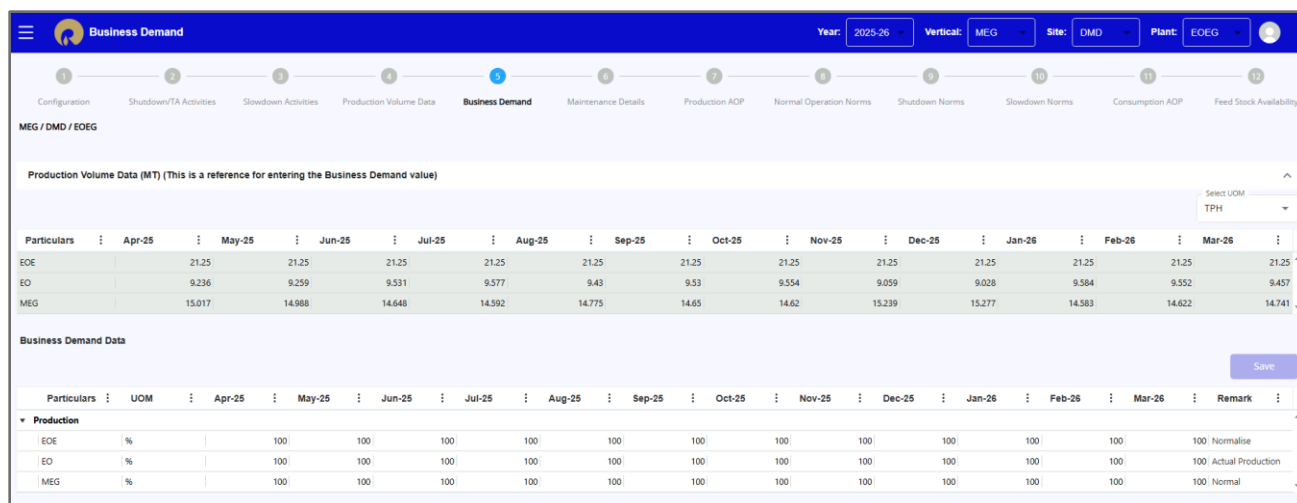


Figure 6 Digitized AOP screen for Business Demand

2.1.1.4 Maintenance/Shutdown/Slowdown/TA Activities

The operations team will provide the shutdown, turnaround and slowdown plan to calculate month-wise operating hours available for production.

The below screenshot depicts the shutdown and turnaround activities planned, as will be captured in the AOP digitized platform.

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration(hrs) to calculate the SD-To date/time.

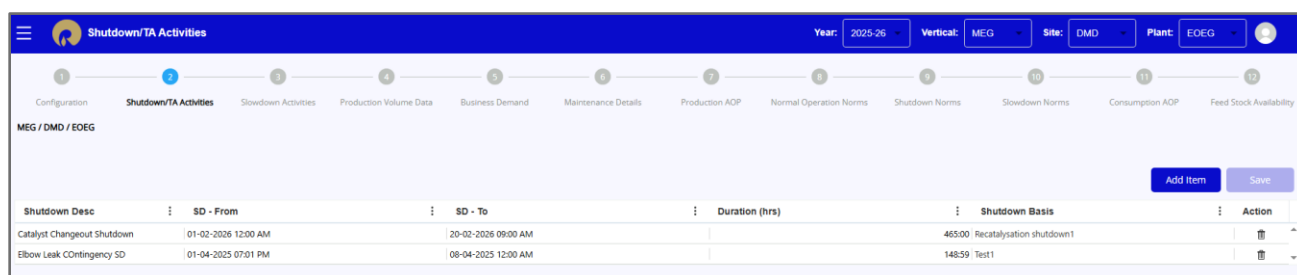


Figure 7 Digitized AOP screen for Shutdown/Turnaround Activities

The Slowdown entries will have the inputs as shutdowns, along with the requirement of plant EOE/EO load (TPH basis) for the duration of the slowdown.

The slowdown activities can be calculated based on the slowdown rate and hours. This is called a Basis Shutdown, equivalent to Shutdown.

$$\text{Slowdown rate(TPH)} = (\text{MCU} - \text{Operating Production rate})$$

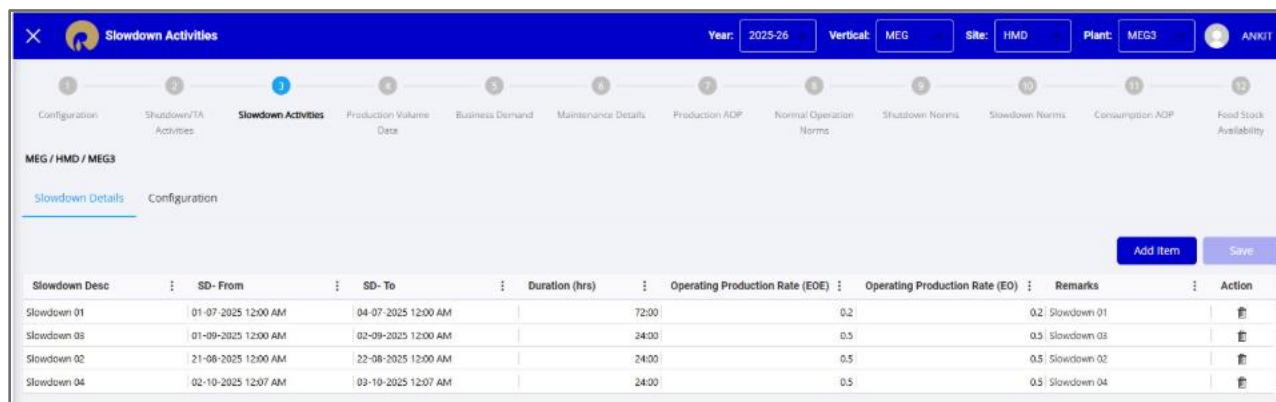
$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate(TPH)} * \text{Slowdown hrs}) / \text{MCU}$$

Operating Production rate will be a manual entry in Slowdown activity screen and slowdown rate will

be calculate at the backend as per above formula.

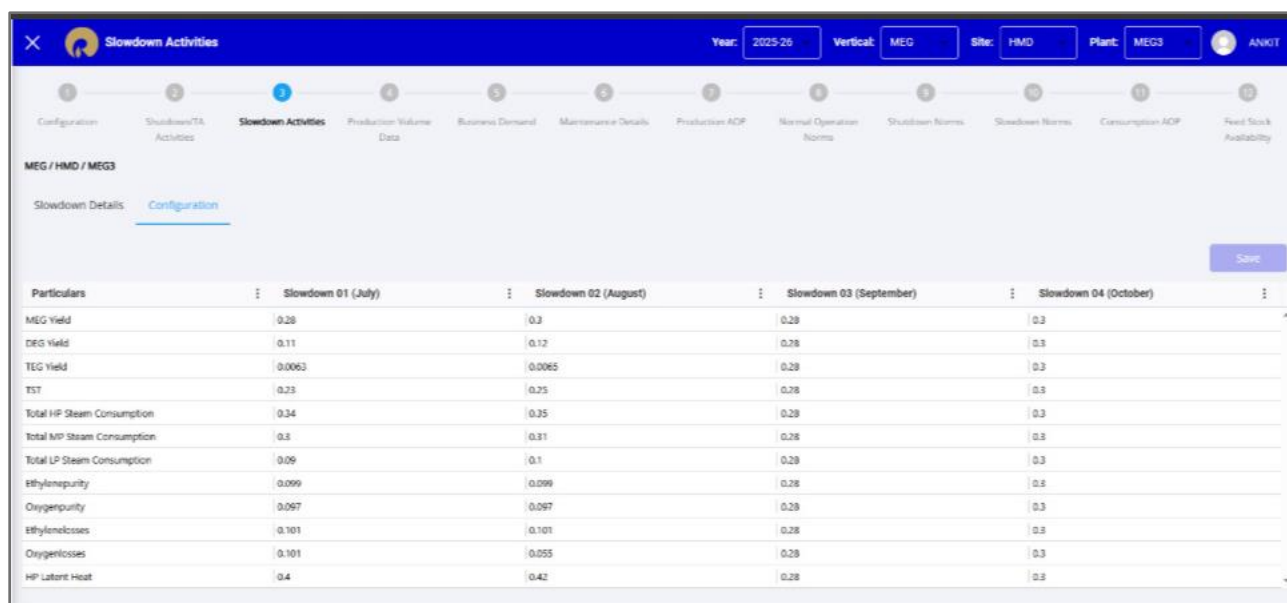
For each shutdown entry, two slowdown entries will be generated automatically, with the nomenclature as “#Shutdown Entry name. – Ramp down” and “#Shutdown Entry name. – Ramp Up”.

In Slowdown details tab, operating production rate-EOE (TPH) and operating production rate-EO (TPH) these two columns will be added in Slowdown activities screen as shown below.



Slowdown Desc	SD- From	SD- To	Duration (hrs)	Operating Production Rate (EOE)	Operating Production Rate (EO)	Remarks	Action
Slowdown 01	01-07-2025 12:00 AM	04-07-2025 12:00 AM	72:00	0.2		0.2 Slowdown 01	 
Slowdown 03	01-09-2025 12:00 AM	02-09-2025 12:00 AM	24:00	0.5		0.5 Slowdown 03	 
Slowdown 02	21-08-2025 12:00 AM	22-08-2025 12:00 AM	24:00	0.5		0.5 Slowdown 02	 
Slowdown 04	02-10-2025 12:07 AM	03-10-2025 12:07 AM	24:00	0.5		0.5 Slowdown 04	 

Separate tab (Configuration) will be provided as shown below in slowdown activity screen for the provision of manual entry related to configuration screen parameters based on the type of slowdown. Number of rows mentioned in slowdown activity screen will be no of column for slow down manual entry screen.



Particulars	Slowdown 01 (July)	Slowdown 02 (August)	Slowdown 03 (September)	Slowdown 04 (October)
MEG Yield	0.28	0.3	0.28	0.3
DEG Yield	0.11	0.12	0.28	0.3
TEG Yield	0.0063	0.0065	0.28	0.3
TST	0.23	0.25	0.28	0.3
Total HP Steam Consumption	0.34	0.35	0.28	0.3
Total MP Steam Consumption	0.3	0.31	0.28	0.3
Total LP Steam Consumption	0.09	0.1	0.28	0.3
Ethylene purity	0.099	0.099	0.28	0.3
Oxygen purity	0.097	0.097	0.28	0.3
Ethylene losses	0.101	0.101	0.28	0.3
Oxygen losses	0.101	0.055	0.28	0.3
HP Latent Heat	0.4	0.42	0.28	0.3

Figure 8 Digitized AOP screen for Slowdown Activities

The 'Add Item' button on the Shutdown/Slowdown activities screen allows users to insert a new entry

for shutdown/slowdown.

The 'Save' button enables users to save newly added items as well as any modifications made to existing entries.

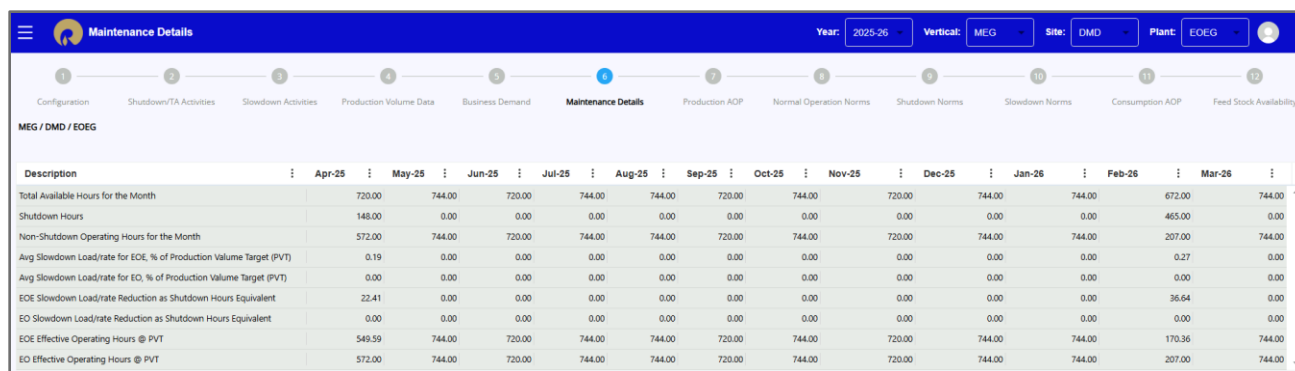
For each month, the following table is generated based on all inputs for shutdowns/slowdowns-

Description	April	...	March
Total Available Hours for the Month	$30 \times 24 = 720$		
Shutdowns Hours	<i>From the Routine Shutdown Plan Section</i>		
Non-Shutdown Operating Hours for the Month	<i>Total Available operating hours for the month – Shutdown hrs</i>		
Average Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EOE \text{ slowdown } hrs_i \times EOE \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EOE \text{ slowdown } hrs_i) \times EOE \text{ MC for Month}}$		
Average Slowdown Load/rate for EO, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EO \text{ slowdown } hrs_i \times EO \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EO \text{ slowdown } hrs_i) \times EO \text{ MC for Month}}$		
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	$EOE \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EOE \text{ slowdown } hrs_i)$		
EO Slowdown Load/rate Reduction as Shutdown Hours Equivalent	$EO \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EO \text{ slowdown } hrs_i)$		
EOE Effective Operating Hours @ PVT	<i>Total Available operating hours for the month – Shutdown/TA hours – EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EOE Slowdown load reduction as Shutdown hours Equivalent (Rampup / RampDown)</i>		
EO Effective Operating Hours @ PVT	<i>Total Available operating hours for the month – Shutdown/TA hours – EO Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EO Slowdown load reduction as Shutdown hours Equivalent (Rampup / RampDown)</i>		

Table 3 Maintenance Details

The screen below depicts the Maintenance Details Table, which is captured in the digitized platform.

NOTE- The below screen is read-only. The users cannot change the values.



Description	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Total Available Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00
Shutdown Hours	148.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	465.00	0.00
Non-Shutdown Operating Hours for the Month	572.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	207.00	744.00
Avg Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	0.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.27	0.00
Avg Slowdown Load/rate for EO, % of Production Volume Target (PVT)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	22.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	36.64	0.00
EO Slowdown Load/rate Reduction as Shutdown Hours Equivalent	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EOE Effective Operating Hours @ PVT	549.59	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	170.36	744.00
EO Effective Operating Hours @ PVT	572.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	207.00	744.00

Figure 9 AOP Maintenance Details Table

2.1.1.5 Calendar Year Based Production Calculations Details/Production AOP

Using the Business Demand, Production Volume data and Maintenance Plan (Operating Hours), the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$\text{Production AOP (MT)} = (\text{Operating hours for EOE/EO in a month (hrs)} * \text{Percentage of product demand} * \text{MC Value of product (TPH)}) / 100$$

$$\text{Operating hours for MEG} = \text{Total Available hours of the month} - \text{Shutdown/TA hours of the month}$$

$$\text{Operating hours for EOE} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

$$\text{Operating hours for EO} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

Hence, month-wise production is,

$$\text{Total Budgeted Production} = \sum \text{All the values for the month}$$

The month-wise budget production is further used to derive the annual plan in KT, and this is part of the final report.

This information is captured in the AOP digitization portal as seen in the screenshot below.

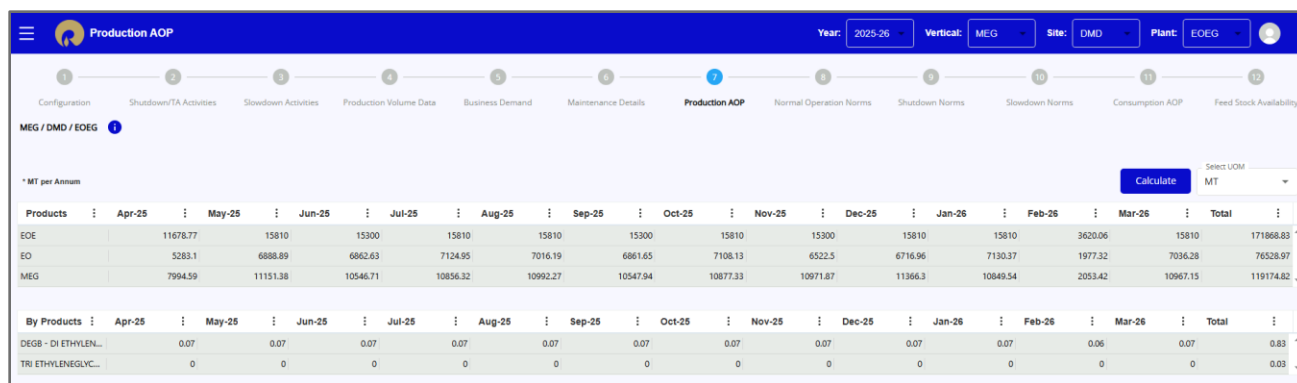


Figure 10 Digitized AOP screen for Production AOP month wise

The '**Calculate**' button on this screen allows users to recalculate Production values when changes are made to configuration settings (e.g., MEG yield), business demand, or when line items are added to slowdown and shutdown activities.

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Normal Operation Norms

All the specific consumption norms are set with One-unit EOE as the basis.

For calculation purpose in the Digitized AOP platform, based on the DCT provided by the user, Norms are broadly classified as –

1. Historical Best
2. Based on Calculations
3. Manual Entries

3.1.2.1.1 Historical Best

Based on the historical data of last 5 years, corresponding to the MCU production, the best-achieved feed slate period is filtered. This data is further used for the calculation of consumption norms, which is the average of the norms during that period, excluding the identified outliers.

Below is the logic formulated to derive the Historical best Normal Operation Norms –

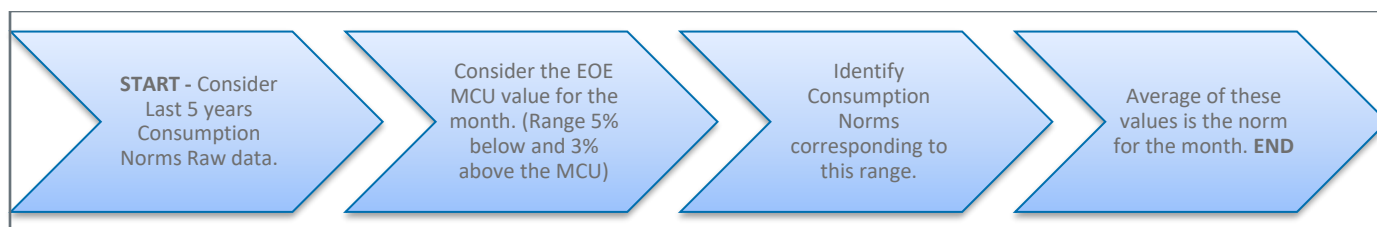


Figure 11 Digitized AOP Steps to derive month wise Historical Best Norms

The user has provision to overwrite/update the value by providing a valid reason for the historical best particulars.

The list of particulars to be derived using the Historical Best logic is specified in the DCT provided by the user. These particulars may vary from plant to plant.

3.1.2.1.2 Based on Calculations

For the MEG vertical, particulars like ethylene, oxygen, HP Steam, BFW water etc. are formulated based on the stoichiometric calculations provided by the user in the DCT.

The formula can vary from plant to plant.

E.g. the Raw Material norms for ethylene and oxygen for DMD EOEG are derived using the calculations mentioned in the below table.

Plant	Category	Particular	Calculation
DMD EOEG	Raw Material	Ethylene	$\frac{(28.05/44.05 \times 100 / \{selectivity\} / 0.9995) + ([METHANE] \times 28.05 \times 0.28 / 16 / 0.54)}{0.9995} + ((1.9 / 24) / ([EOE] / [EOE \text{ Effective Op hrs}]))$ Where, 28.05 - Ethylene Molecular Weight 44.05 - EO molecular weight 0.9995 - Ethylene Purity (wt%) 0.28 - Ethylene concentration in argon purge (Mol %) 16 - Methane molecular weight 0.55 - Methane concentration in argon purge (Mol %) 1.9 - Allocated Ethylene consumptions (TPD)
		Oxygen	$\frac{(32/44.05 \times (3 / \{selectivity\} \times 100 - 2.5) / 0.996) + ([METHANE] \times 32 \times 0.052 / 16 / 0.54)}{0.996}$ Where, 32 - Oxygen molecular weight 44.05 - EO molecular weight 3,2.5 - Stoichiometric Ratios 0.996 - Oxygen Purity (wt%) 0.052 - Oxygen concentration in argon purge (Mol %) 16 - Methane molecular weight 0.54 - Methane concentration in argon purge (Mol %)

Table 4 Raw Material Norms calculations

The values for *losses and purity* can be captured from the Lab analysis or Past/last year's operations. The user has the provision to enter the value manually in the Configuration screen.

Selectivity, Losses and Purity can be modified/added by the user in the Configuration Screen under the Calculation Input for Norms Subsection.

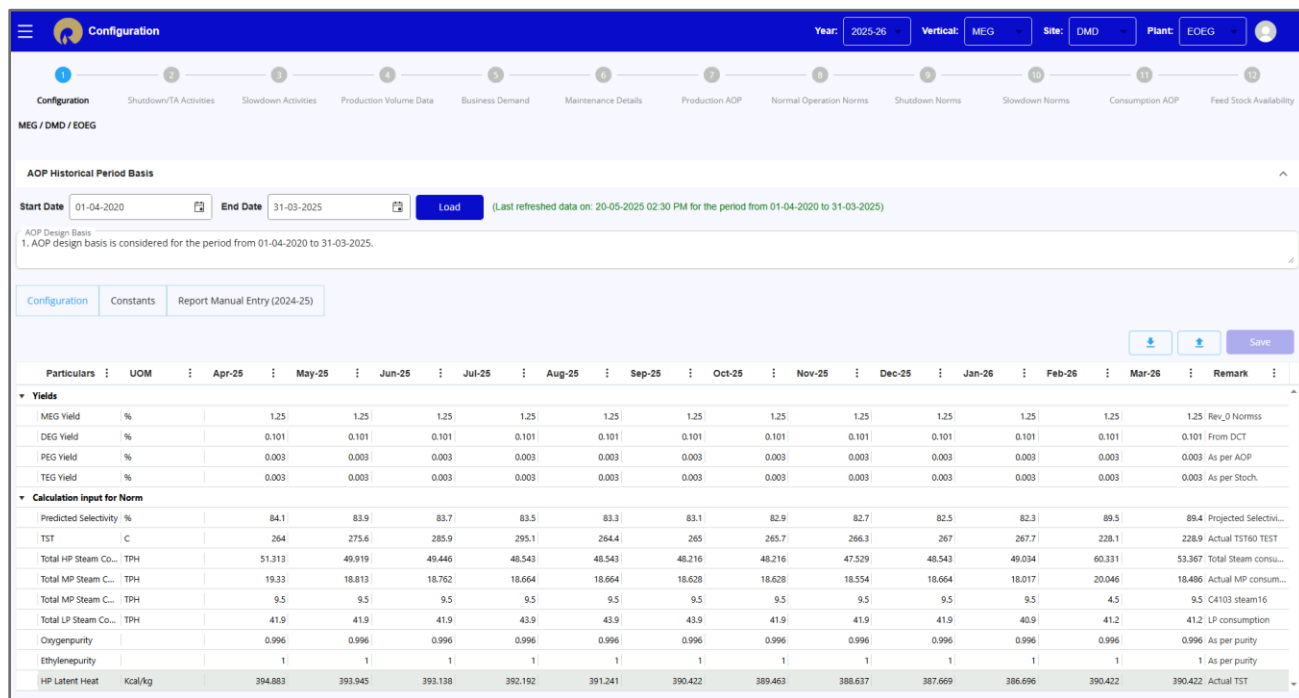


Figure 12 Digitized AOP screen for Catalyst Selectivity month wise

In the Normal Operations Norms screen the cells for particulars with formula, e.g. HP Steam Dis, will be read only. If the values for such particulars are to be modified then the respective inputs should be modified, e.g. Total HP Steam consumption in the Configuration Screen.

3.1.2.1.3 Manual Entries

The particulars under the Chemicals and Catalyst sections are manually entered by the user, who must also provide the corresponding norm values in the Normal Operation Norms screen.

The screenshot below highlights the details captured in the digitized AOP platform in the Normal Operations Norms screen.

Figure 13 Digitized AOP screen for Normal operations Norms

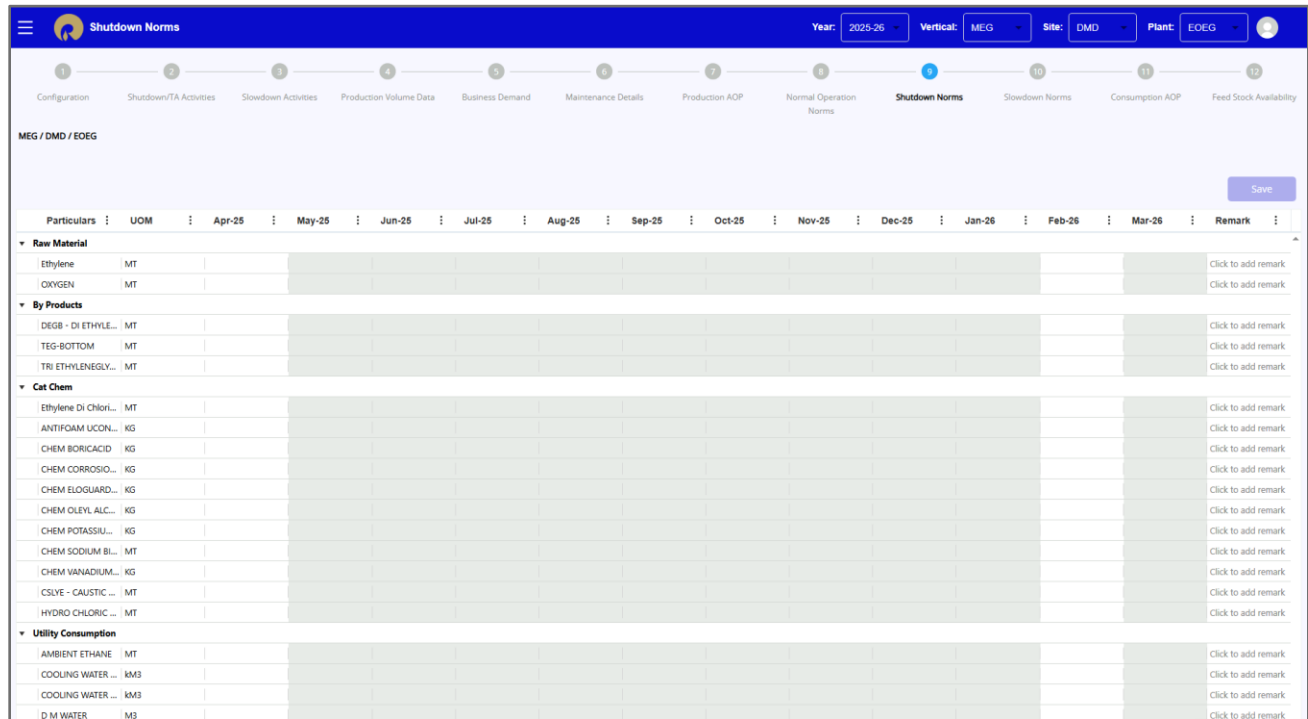
Intermediate Values														
Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
▼ Calculated Intermediate Values														
Excess:HP	TPH	0	0	0	0	0	0	0	0	0	0	0	0	0
Excess:MP	TPH	3.6453	4.5208	4.8805	5.3373	5.6967	6.0418	6.4012	6.7847	7.0347	8.0422	0	0	0
Heat.to:HP.Steam	kcal/h	29796400	21239800	21528400	21795800	22065100	22359800	22663300	22932100	23209500	23489000	17946900	17951100	
Heat.to:MP.Steam	kcal/h	10874200	11043900	11190000	11359800	11529900	11676200	11846300	11992800	12163200	12333800	2858580	3047970	
HP.Steam.Generation	TPH	42.7069	43.261	43.8647	44.4286	44.9974	45.6159	46.195	46.8236	47.4126	48.0069	39.7557	39.6912	
MP.Steam.Generation	TPH	22.3753	23.338	23.6425	24.0013	24.3607	24.6698	25.0292	25.3387	25.6987	26.0592	6.03968	6.43983	
Reaction:Heat	kcal/h	41164300	41596600	42031000	42467500	42906100	43346800	43789700	44234700	44681800	45131100	30220800	30411550	

The **'Save'** button on the above screen enables the user to save any norm value updated manually for historical best norms for cat chem.

The shutdown quantity utilization data includes TA, asset maintenance linked shutdown/ slowdown activities, etc..

The user must enter the consumption data (absolute values) for each line item.

The below screenshot depicts the Shutdown Norms captured in the AOP digitized platform.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT													Click to add remark
OXYGEN	MT													Click to add remark
By Products														
DEGB - DI ETHYLE...	MT													Click to add remark
TEG-BOTTOM	MT													Click to add remark
TRI ETHYLENEGLY...	MT													Click to add remark
Cat Chem														
Ethylene Di Chlori...	MT													Click to add remark
ANTIFOAM UCON...	KG													Click to add remark
CHEM BORICACID	KG													Click to add remark
CHEM CORROSIO...	KG													Click to add remark
CHEM ELOGUARD...	KG													Click to add remark
CHEM OLEYL ALC...	KG													Click to add remark
CHEM POTASSIU...	KG													Click to add remark
CHEM SODIUM BL...	MT													Click to add remark
CHEM VANADIUM...	KG													Click to add remark
CSLYE - CAUSTIC ...	MT													Click to add remark
HYDRO CHLORIC ...	MT													Click to add remark
Utility Consumption														
AMBIENT ETHANE	MT													Click to add remark
COOLING WATER ...	km3													Click to add remark
COOLING WATER ...	km3													Click to add remark
D M WATER	M3													Click to add remark

Figure 14 Digitized AOP screen for Shutdown Norms

2.1.2.3 Slowdown Norms

Slowdown norms are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The values for slowdown norms are same as the Normal Operations norms for the month.

NOTE- Slowdown wise norms calculation will be provided by RIL.

The below screenshot depicts the Shutdown Norms captured in the AOP digitized platform.

MEG / DMD / EOEG

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT	0.76633	0	0	0	0	0	0	0	0	0	0.72084	0	Updated by system
OXYGEN	MT	0.77875	0	0	0	0	0	0	0	0	0	0.62179	0	Updated by system
By Products														
DEGB - DI ETHYLE...	MT	0.06914	0	0	0	0	0	0	0	0	0	0.05729	0	Updated by system
TEG-BOTTOM	MT	0.00205	0	0	0	0	0	0	0	0	0	0.0017	0	Updated by system
TRI ETHYLENEGLY...	MT	0.00226	0	0	0	0	0	0	0	0	0	0.00187	0	Updated by system
Cat Chem														
Ethylene DI Chlori...	MT	0.0265	0	0	0	0	0	0	0	0	0	0.0265	0	Updated by system
ANTIFOAM UCON...	KG	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM BORICACID	KG	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM CORROSIQ...	KG	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM ELOGUARD...	KG	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM OLEYL AL...	KG	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM POTASSIU...	KG	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM SODIUM BL...	MT	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CHEM VANADIUM...	KG	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
CSLVE - CAUSTIC ...	MT	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
HYDRO CHLORIC ...	MT	0	0	0	0	0	0	0	0	0	0	0	0	Updated by system
Utility Consumption														
AMBIENT ETHANE	MT	0.0001	0	0	0	0	0	0	0	0	0	0.00034	0	Updated by system
COOLING WATER ...	KM3	0.4706	0	0	0	0	0	0	0	0	0	0.48904	0	Updated by system
COOLING WATER ...	KM3	0.07436	0	0	0	0	0	0	0	0	0	0.06285	0	Updated by system

Figure 15 Digitized AOP screen for Slowdown Norms

2.1.2.4 Budget Norms/Consumption AOP

Using the values of Normal operations Norms, Shutdown Consumptions and Slowdown Norms, the final Consumption AOP value is calculated as below –

Consumption AOP for months without shutdown or slowdown = Normal Operations Norms

*Consumption AOP for months with shutdown or slowdown = Total Consumption/
Total production*

Where,

*Total Consumption = (Normal Operation Norms * EOE MC value * EOE operating hours) +
 $\sum_{i=1}^n$ (Slowdown Norms * EOE slowdown hours * EOE slowdown rate)_n + Shutdown Consumption*

*Total Production = (EOE MC value * EOE operating hours) + $\sum_{i=1}^n$ (EOE slowdown hours *
EOE slowdown rate)_n*

The below screenshot depicts the Shutdown Norms captured in the AOP digitized platform.

Consumption AOP

Year:

2025-26

Vertical:

MEG

Site:

DMD

Plant:

EOEG

1

2

3

4

5

6

7

8

9

10

11

12

Configuration

Shutdown/TA Activities

Slowdown Activities

Production Volume Data

Business Demand

Maintenance Details

Production AOP

Normal Operation Norms

Shutdown Norms

Slowdown Norms

Consumption AOP

Feed Stock Availability

MEG / DMD / EOEG

Calculate

Particulars	UOM / MT	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Raw Material													
Ethylene	MT	0.76608	0.76816	0.77075	0.7723	0.77438	0.77634	0.77758	0.77941	0.78078	0.78284	0.7206	0.7217
OXYGEN	MT	0.77944	0.7857	0.7921	0.7983	0.80465	0.81099	0.81721	0.82359	0.8299	0.83638	0.62251	0.62531
By Products													
DEGB - DI ETHYLEN...	MT	0.07495	0.07497	0.07334	0.07299	0.07398	0.07336	0.07313	0.07591	0.07612	0.073	0.07312	0.07367
TEG-BOTTOM	MT	0.00223	0.00223	0.00218	0.00217	0.0022	0.00218	0.00217	0.00225	0.00226	0.00217	0.00217	0.00219
TRI ETHYLENEGLYC...	MT	0.00245	0.00245	0.0024	0.00238	0.00242	0.0024	0.00239	0.00248	0.00249	0.00238	0.00239	0.00241
Cat Chem													
Ethylene Di Chloride	MT	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265	0.0265
ANTIFOAM UCON-...	KG												
CHEM BORICACID	KG												
CHEM CORROSION...	KG												
CHEM ELOGUARD86	KG												
CHEM OLEYL ALCO...	KG												
CHEM POTASSIUM ...	KG												
CHEM SODIUM BIS...	MT												
CHEM VANADIUM ...	KG												
CSLYE - CAUSTIC S...	MT												
HYDRO CHLORIC A...	MT												
Utility Consumption													
AMBIENT ETHANE	MT	0.0001	0.0001	0.00001	0	0.00007	0.00031	0.00005	0.00031	0.00042	0.00048	0.00034	0.00056
COOLING WATER D4	KM3	0.4706	0.49025	0.49775	0.49976	0.49456	0.48956	0.49039	0.49011	0.48807	0.49	0.48904	0.47625
COOLING WATER D5	KM3	0.07436	0.08069	0.08083	0.08629	0.08605	0.07846	0.07561	0.08193	0.08112	0.08206	0.08285	0.07352

Figure 16 Digitized AOP screen for Budget Norms or Consumption Norms

2.1.3 Feedstock Availability

Availability of Feedstock/Raw Material defines maximum attainable production in a plant. In the case of MEG technology, the availability of Ethylene and Oxygen will be accounted for planning of unit production.

For the Rev 0 for AOP production/norms will consider the 100% availability of the required.

- These quantities will be compared against feedstock requirements and used for optimizing the production to meet the constrained feedstock availability as per below mentioned logic.

Reduced No of Operating Hrs to Maximize the production by making Delta Feedstock (difference of Available and Required feedstock) = 0

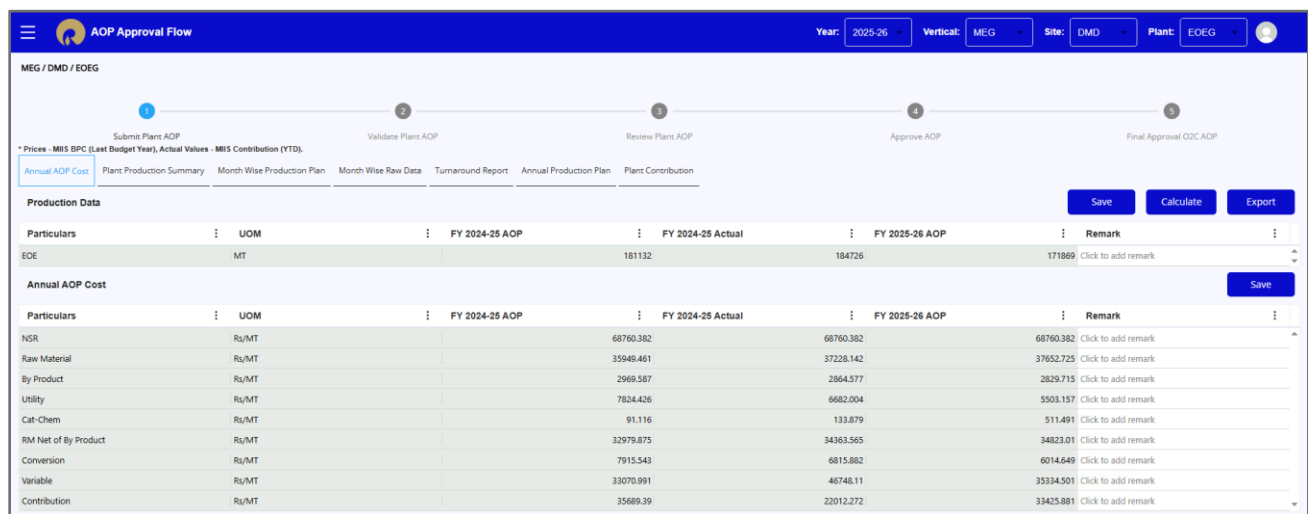
2.1.4 Other Cost (To generate cost sheet - Business)

- Process consumables will be discussed with RIL and TBD for a path forward.
- Maintenance consumables will be discussed with RIL and TBD for a path forward.

3. AOP Reports

- Dataset (Basis)
- Annual AOP Cost.
- Plant Production Summary (T-14).
- Month wise Production Plan (T – 16)
- Month wise Raw data (T – 18).
- Turnaround Report (T – 19A).
- Annual Production Plan (T – 15).
- Plant Contribution (T – 21).

The below screenshot depicts the AOP report captured in the AOP digitized platform.



The screenshot shows the 'AOP Approval Flow' interface. At the top, there are filters for Year (2025-26), Vertical (MEG), Site (DMD), and Plant (EOEG). Below this is a progress bar with five steps: 1. Submit Plant AOP, 2. Validate Plant AOP, 3. Review Plant AOP, 4. Approve AOP, and 5. Final Approval Q2C AOP. The current step is 3, 'Review Plant AOP'. Below the progress bar, there are tabs for 'Annual AOP Cost', 'Plant Production Summary', 'Month Wise Production Plan', 'Month Wise Raw Data', 'Turnaround Report', 'Annual Production Plan', and 'Plant Contribution'. The 'Annual AOP Cost' tab is selected. The main table displays 'Production Data' with columns: Particulars, UOM, FY 2024-25 AOP, FY 2024-25 Actual, FY 2025-26 AOP, and Remark. The table includes rows for 'EOE' and 'Annual AOP Cost' with various sub-items like NSR, Raw Material, By Product, Utility, Cat-Chem, RM Net of By Product, Conversion, Variable, and Contribution. Each row has a 'Save' button next to it.

Particulars	UOM	FY 2024-25 AOP	FY 2024-25 Actual	FY 2025-26 AOP	Remark
EOE	MT	181132		184726	171869 Click to add remark
Annual AOP Cost					
NSR	Ru/MT	68760.382	68760.382	68760.382	68760.382 Click to add remark
Raw Material	Ru/MT	35949.461	37228.142	37652.725	37652.725 Click to add remark
By Product	Ru/MT	2969.587	2864.577	2829.715	2829.715 Click to add remark
Utility	Ru/MT	7824.426	6682.004	5503.157	5503.157 Click to add remark
Cat-Chem	Ru/MT	91.116	133.879	511.491	511.491 Click to add remark
RM Net of By Product	Ru/MT	32979.875	34363.565	34823.01	34823.01 Click to add remark
Conversion	Ru/MT	7915.543	6815.882	6014.649	6014.649 Click to add remark
Variable	Ru/MT	33070.991	46748.11	35334.501	35334.501 Click to add remark
Contribution	Ru/MT	35689.39	22012.272	33425.881	33425.881 Click to add remark

Figure 17 Digitized AOP screen for Annual AOP Cost

AOP Approval Flow

Year:

2025-26

Vertical:

MEG

Site:

DMD

Plant:

EOEG

MEG / DMD / EOEG

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval Q2C AOP

* Prices - M1S BPC (Last Budget Year), Actual Values - M1S Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Plant Production Summary (T-14)

Save

SL.No	Item	Unit	2024-25		2025-26		Variance wrt current year budget		Variance wrt current year actuals		Remarks
	Production Volume		Budget	Actual	Budget	MT	%	MT	%		
1	EOE	MT	181131.81	184726.313	171868.83	-9262.98	-5.114	-12857.483	-6.96	Click to add remark	
2	MEG	MT	138201.238	132761.817	119174.821	-19026.417	-13.767	-13386.996	-10.234	Click to add remark	
3	EO	MT	70570.82	78393.985	76528.973	5958.153	8.443	-1865.012	-2.379	Click to add remark	
4	Production	MT	181131.81	184726.313	171868.83	-9262.98	-5.114	-12857.483	-6.96	Click to add remark	
5	Operating Hours	Hrs	8760	8760	8147	-613	-6.998	-613	-6.998	Click to add remark	
6	ProductionRate	TPH	20.677	21.087	21.25	0.573	2.77	0.163	0.771	Click to add remark	
7	NSR	Rs/MT	68760.382	68760.382	68760.382	0	0	0	0	Click to add remark	
8	Raw Material	Rs/MT	35949.461	37228.142	37652.725	-1703.263	-4.738	-424.583	-1.14	Click to add remark	
9	By Product	Rs/MT	2969.587	2864.377	2829.715	-139.871	-4.71	-34.862	-1.217	Click to add remark	
10	RM Net of By Product	Rs/MT	32979.875	34363.565	34823.01	-1843.135	-5.589	-459.445	-1.337	Click to add remark	
11	Cat-Chem	Rs/MT	91.116	133.879	511.491	-420.375	-461.361	-377.613	-282.056	Click to add remark	
12	Utility	Rs/MT	7824.426	6682.004	5503.157	2321.269	29.667	1178.846	17.642	Click to add remark	
13	Conversion	Rs/MT	7915.543	6615.882	6014.649	1900.894	24.015	801.233	11.755	Click to add remark	
14	Variable	Rs/MT	33070.991	46748.11	35334.501	-2263.51	-6.844	11413.609	24.415	Click to add remark	
15	Contribution	Rs/MT	35689.39	22012.272	33425.881	-2263.51	-6.342	11413.609	51.851	Click to add remark	
16	Contribution Rs Lacs	Rs lacs	64644.839	40662.458	57448.67	-7196.169	-11.132	16786.212	41.282	Click to add remark	

Figure 18 Digitized AOP screen for Production Plant Summary

AOP Approval Flow

Year:

2025-26

Vertical:

MEG

Site:

DMD

Plant:

EOEG

MEG / DMD / EOEG

1

Submit Plant AOP

2

Validate Plant AOP

3

Review Plant AOP

4

Approve AOP

5

Final Approval Q2C AOP

* Prices - MIS BPC (Last Budget Year), Actual Values - MIS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Production (T-16)

Save

SL...	Month	2024-25						2025-26						Remarks
		EOE Production, MT		Operating Hours		Throughput, TPH		Operating ...	MEG Throu...	EO Throu...	EOE Throu...	Total EOE, ...		
		Budget	Actual	Budget	Actual	Budget	Actual							
1	April	11872	16259	720	720	16.49	22.58	572	15.02	9.24	21.25	12155	Click to add remark	
2	May	16864	15527	744	744	22.67	20.87	744	14.99	9.26	21.25	15810	Click to add remark	
3	June	16320	15538	720	720	22.67	21.58	720	14.65	9.53	21.25	15300	Click to add remark	
4	July	16492	16138	744	744	22.17	21.69	744	14.59	9.58	21.25	15810	Click to add remark	
5	August	15996	15915	744	744	21.5	21.39	744	14.77	9.43	21.25	15810	Click to add remark	
6	September	16864	15261	720	720	6.51	21.2	720	14.65	9.53	21.25	15300	Click to add remark	
7	October	16864	15730	744	744	22.67	21.14	744	14.62	9.55	21.25	15810	Click to add remark	
8	November	16236	14178	720	720	22.55	19.69	720	15.24	9.06	21.25	15300	Click to add remark	
9	December	16864	14649	744	744	22.67	19.69	744	15.28	9.03	21.25	15810	Click to add remark	
10	January	16864	15589	744	744	22.67	20.95	744	14.58	9.58	21.25	15810	Click to add remark	
11	February	15232	14347	672	672	22.67	21.35	207	14.62	9.55	21.25	4396.75	Click to add remark	
12	March	16844	15594	744	744	22.64	20.96	744	14.74	9.46	21.25	15810	Click to add remark	
13	Total	181132	184725	8760	8760	20.68	21.09	8147	14.82	9.39	21.25	173123.75	Click to add remark	

Main Products - Production for the budget year

Products	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
EOE	11678.77	15810	15300	15810	15810	15300	15810	15300	15810	15810	3620.06	15810	171868.83
EO	5283.1	6888.89	6862.63	7124.95	7016.19	6861.65	7108.13	6522.5	6716.96	7130.37	1977.32	7036.28	76528.97
MEG	7994.59	11151.38	10546.71	10856.32	10992.27	10547.94	10877.33	10971.87	11366.3	10849.54	2053.42	10967.15	119174.82

Figure 19 Digitized AOP screen for Month Wise Production Plan

AOP Approval Flow

Year: 2025-26Vertical: MEGSite: DMDPlant: EOE

MEG / DMD / EOE

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval QIC AOP

* Prices - MBS BPC (Last Budget Year), Actual Values - MBS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Consumption (T-18)

Parameters	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Budgeted Selectivity L...	88.6	88.4	88.2	88	87.8	87.6	87.4	87.2	87	86.8	86.6	86.4
Actual Selectivity for L...	89.22	89	88.77	88.68	88.49	88.34	88.16	87.93	87	86.8	86.6	86.4
Guaranteed Selectivity...	86.2	85.9	85.6	85.3	85	84.7	84.3	84	83.6	83.2	82.8	82.4
Predicted Selectivity L...	84.1	83.9	83.7	83.5	83.3	83.1	82.9	82.7	82.5	82.3	82.1	81.9

RawMaterial

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
OXYGEN	MT	0.787806	9102.889	12421.949	12119.069	12621.107	12721.437	12408.178	12920.138	12600.942	13120.719	13223.215	2253.516	9886.088	135399.247
Ethylene	MT	0.769034	8946.895	12144.625	11792.49	12210.079	12242.9	11878.017	12293.524	11924.988	12344.195	12376.685	2608.63	11410.014	132173.043

ByProducts

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
TEG-BOTTOM	MT	0.00220062	25.998	35.204	33.332	34.279	34.741	33.341	34.341	34.497	35.748	34.279	7.862	34.595	378.218
TRI ETHYLENEGL...	MT	0.00242068	28.598	38.725	36.665	37.707	38.215	36.675	37.775	37.947	39.323	37.707	8.648	38.055	416.039
DI ETHYLENE GLY...	MT	0.0740874	875.277	1185.206	1122.165	1154.048	1169.619	1122.475	1156.147	1161.403	1203.533	1154.054	264.688	1164.705	12733.321

CatChem

Particul...	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
EDC	MT	0.0265	309.487	418.965	405.45	418.965	418.965	405.45	418.965	405.45	418.965	418.965	95.932	418.965	4554.524
HYDRO CHLORIC...	MT														
ANTIFOAM UCO...	KG														
CAUSTIC SODA L...	MT														
CHEM BORICACID	KG														
CHEM CORROSI...	KG														

Figure 20 Digitized AOP screen for Month Wise Raw Data

AOP Approval Flow														Year: 2025-26	Vertical: MEG	Site: DMD	Plant: EOE	
MEG / DMD / EOE																		
1 2 3 4 5 Submit Plant AOP Validate Plant AOP Review Plant AOP Approve AOP Final Approval QIC AOP * Prices - MBS BPC (Last Budget Year), Actual Values - MBS Contribution (YTD). Annual AOP Cost Plant Production Summary Month Wise Production Plan Month Wise Raw Data Turnaround Report Annual Production Plan Plant Contribution																		
Turnaround Details (T-19A)																		
SL.No Activities Turnaround Period From To Duration, hrs Remark 1 Elbow Leak Contingency SD 01-Apr-25 08-Apr-25 168.00 Click to add remark 2 Elbow Leak Contingency SD 01-Apr-25 08-Apr-25 148.00 Click to add remark 3 Catalyst Changeout Shutdown 01-Feb-26 20-Feb-26 465.00 Click to add remark																		
Turnaround details for the previous years data since commissioning																		
SL.No Activities Turnaround Period From Turnaround Period To Duration, hrs Period in Months Remark Action 1 Elbow Leak Contingency SD 01-Apr-25 20-Feb-26 633 Click to add remark 2 Elbow Leak Contingency SD 01-Apr-25 20-Feb-26 613 Click to add remark																		

Figure 21 Digitized AOP screen for Turnaround Report

Figure 22 Digitized AOP screen for Annual Production Plan

Figure 23 Digitized AOP screen for Plant Contribution

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by DMD Site team